

Sampled Coefficient Contraction and Nondivergence

Collatz proof project

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Abstract

We prove in Lean that universal coefficient contraction observed only at visits to two modulo nine is equivalent to boundedness of every positive Collatz orbit. The proof selects inverse-coefficient maxima on the sampled visits and controls intervening coefficients by a factor of sixty-four. Neither equivalent statement is proved here. Nontrivial cycles are not excluded, and no mathematical priority claim is made.

Let $n_t = T^t(n)$ for the shortcut Collatz map, let $j_t(n)$ count odd steps, and write $C_t(n) = 3^{j_t(n)}/2^t$. Call n sampled-noncontracting if $n \equiv 2 \pmod{9}$ and $C_t(n) \geq 1$ whenever $n_t \equiv 2 \pmod{9}$.

Theorem 1. *Every sampled-noncontracting seed satisfies $C_t(n) \geq 1/64$ at all times. Every positive sampled-noncontracting seed has an unbounded orbit.*

Proof. Fix a time t and take the last residue-two visit $k \leq t$; time zero is such a visit. Our previous residue-height theorem bounds the number of even steps between k and t by six. The coefficient on that segment is therefore at least $1/64$. Multiplying by $C_k(n) \geq 1$ gives the bound.

Suppose all orbit values were at most B . The exact affine identity $2^t n_t = 3^{j_t(n)} n + d_t(n)$ then gives $d_t(n)/3^{j_t(n)} \leq 64B$ for all t . This bounds the ideal correction, which our earlier formal summability theory proves equivalent to escape for positive seeds. This contradicts the assumed orbit bound. $\square$

Theorem 2. *Every unbounded positive orbit has a sampled-noncontracting tail.*

Proof. First shift to a visit to two modulo nine, using formal forward coverage. The shifted orbit is still unbounded. Its inverse coefficient $f(t) = 1/C_t$ tends to zero by the existing summability theorem, and $f(0) = 1$. Consequently f attains a maximum on the subset of times whose orbit value is two modulo nine: a sufficiently late tail has $f(t) < 1$, so only finitely many earlier sampled times need be compared. Choose a maximizing sampled time s . For every subsequent sampled visit at time $s+t$, the coefficient cocycle gives $1/C_t(n_s) = f(s+t)/f(s) \leq 1$. Thus n_s is the required seed. $\square$

Theorem 3. *Every positive orbit is bounded if and only if for every positive $n \equiv 2 \pmod{9}$ some t satisfies $n_t \equiv 2 \pmod{9}$ and $C_t(n) < 1$.*

Proof. Under universal boundedness, failure of sampled contraction contradicts Theorem 1. Conversely an unbounded orbit supplies, by Theorem 2, a positive sampled-noncontracting seed, contradicting universal sampled contraction. $\square$

Scope and remaining work. This resolves simultaneous selection of a residue and *sampled* noncontraction, not selection with full noncontraction at every time. The latter property cannot be substituted for the former in a proof. Universal sampled contraction is still an open nondivergence obligation. The separate candidate that the first sampled contraction forces descent also remains unproved; the earlier finite scan supplies evidence only. Neither the equivalence nor its unproved sides exclude nontrivial cycles.

Formal artifacts. The module `Collatz.SampledContraction` contains the definition `SampledNeverContracts`, its `deficit` and `unbounded` theorems, `unbounded_has_sampled_noncontracting_tail`, and `all_bounded_iff_sampled_contraction`. It imports the project's proved residue-coverage and noncontracting-tail results. No new axiom or experimental premise is used.